

Q1 Reverse a list

1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i;
5     printf("enter the number of elements: ");
6     scanf("%d",&n);
7
8     int arr[n];
9     printf("enter %d elements:\n",n);
10    for(i=0;i<n;i++)
11    {
12        scanf("%d",&arr[i]);
13    }
14    printf("array in reverse order:\n");
15    for(i=n-1;i>=0;i--)
16    {
17        printf("%d ",arr[i]);
18    }
19    return 0;
20 }
```

 Submitted

Input

5
10 20 30 40 50

Output

enter the number of elements: enter
5 elements:
array in reverse order:
50 40 30 20 10

Q2 Reverse a string

1 mark

Write a C program for palindrom

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     char str[100], rev[100];
5     int i = 0, len = 0;
6
7     printf("Enter a strings: ");
8     scanf("%99s", str);
9
10    while (str[len] != '\0') {
11        len++;
12    }
13
14    for (i = 0; i < len; i++) {
15        rev[i] = str[len - 1 - i];
16    }
17
18    rev[len] = '\0';
19
20    for (i = 0; i < len; i++) {
21        if (str[i] != rev[i]) {
22            printf("Not a palindrome");
23            return 0;
24        }
25    }
26
27    printf("palindrome");
28    return 0;
29 }
```

Submitted

Input

madam

Output

Enter a string: palindrome

ANJALI OVIYA U (192521452)

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int target) {
4     int left = 0, right = n - 1;
5
6     while (left <= right) {
7         int mid = left + (right - left) / 2;
8
9         if (arr[mid] == target)
10             return mid; // element found
11         else if (arr[mid] < target)
12             left = mid + 1;
13         else
14             right = mid - 1;
15     }
16
17     return -1; // element not found
18 }
19
20 int main() {
21     int n, target;
22
23     printf("Enter number of elements: ");
24     scanf("%d", &n);
25
26     int arr[n];
27     printf("Enter sorted elements: ");
28     for (int i = 0; i < n; i++) {
29         scanf("%d", &arr[i]);
30     }
31
32     printf("Enter element to search: ");
33     scanf("%d", &target);
```

Input

```
5
10 20 30 40 50
30
```

Output

ANJALI OVIYA U (192521452)

Q4 Searching

1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, x, count = 0;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     int arr[n];
10    printf("Enter elements: ");
11    for (i = 0; i < n; i++) {
12        scanf("%d", &arr[i]);
13    }
14
15    printf("Enter element to count: ");
16    scanf("%d", &x);
17
18    for (i = 0; i < n; i++) {
19        if (arr[i] == x)
20            count++;
21    }
22
23    printf("Occurrence of %d is %d\n", x, count);
24    return 0;
25 }
```

Submitted

Input

```
5
1 4 3 4 2
4
```

Output

```
Enter number of elements: Enter
elements: Enter element to count:
Occurrence of 4 is 2
```

ANJALI OVIYA U (192521452)

Write a program for binary search

Your Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int target) {
4     int low = 0, high = n - 1;
5
6     while (low <= high) {
7         int mid = low + (high - low) / 2;
8
9         if (arr[mid] == target)
10             return mid;
11         else if (arr[mid] < target)
12             low = mid + 1;
13         else
14             high = mid - 1;
15     }
16
17     return -1;
18 }
19
20 int main() {
21     int n;
22     printf("Enter number of elements: ");
23     scanf("%d", &n);
24
25     int arr[n];
26     printf("Enter sorted elements: ");
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &arr[i]);
29     }
30
31     int target;
32     printf("Enter element to search: ");
33     scanf("%d", &target);
```

Input

```
5
1 3 5 7 9
5
```

Output

```
Enter number of elements: Enter
sorted elements: Enter element to
search: Element found at index 2
```

ANJALI OVIYA U (192521452)

Write a C program for Merge Two Arrays

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n1, n2;
5
6     printf("Enter size of first array: ");
7     scanf("%d", &n1);
8
9     int a[n1];
10    printf("Enter elements of first array (sorted): ");
11    for (int i = 0; i < n1; i++) scanf("%d", &a[i]);
12
13    printf("Enter size of second array: ");
14    scanf("%d", &n2);
15
16    int b[n2];
17    printf("Enter elements of second array (sorted): ");
18    for (int i = 0; i < n2; i++) scanf("%d", &b[i]);
19
20    int merged[n1 + n2];
21    int i = 0, j = 0, k = 0;
22
23    // Merge
24    while (i < n1 && j < n2) {
25        if (a[i] <= b[j]) merged[k++] = a[i++];
26        else merged[k++] = b[j++];
27    }
28
29    // Copy remaining elements
30    while (i < n1) merged[k++] = a[i++];
31    while (j < n2) merged[k++] = b[j++];
32
33    printf("Merged array: ");
```

Input

```
5
1 3 5 7 9
4
2 4 6 8
```

Output

```
Enter size of first array: Enter
elements of first array (sorted):
Enter size of second array: Enter
elements of second array (sorted):
Merged array: 1 2 3 4 5 6 7 8 9
```

ANJALI OVIYA U (192521452)

Q7 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int arr[100];
6     int smallest, largest;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11     printf("Enter %d elements: ", n);
12     for (i = 0; i < n; i++) {
13         scanf("%d", &arr[i]);
14     }
15
16     smallest = arr[0];
17     largest = arr[0];
18
19     for (i = 1; i < n; i++) {
20         if (arr[i] < smallest)
21             smallest = arr[i];
22         if (arr[i] > largest)
23             largest = arr[i];
24     }
25
26     printf("Smallest element: %d\n", smallest);
27     printf("Largest element: %d\n", largest);
28
29     return 0;
30 }
```

Input

```
6
3 5 1 9 8 4
```

Output

[Run](#) [Save](#) [Submit](#)

ANJALI OVIYA U (192521452)

Second Largest Element

1 mark

Write a C program for Find the Second Largest Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     printf("Enter number of elements: ");
6     scanf("%d", &n);
7
8     int a[n];
9     printf("Enter %d elements: ", n);
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &a[i]);
12    }
13
14
15    int largest = a[0];
16    int second_largest = -2147483648;
17
18    for (int i = 0; i < n; i++) {
19
20        if (a[i] == largest) continue;
21
22        if (a[i] > largest) {
23            second_largest = largest;
24            largest = a[i];
25        } else if (a[i] > second_largest) {
26            second_largest = a[i];
27        }
28    }
29
30    if (second_largest == -2147483648) {
31        printf("Second largest element does not exist (all elements may be same)");
32    } else {
33        printf("Second largest element = %d\n", second_largest);
34    }
35 }
```

Input

5
1 3 4 2 5

Output

Enter number of elements: Enter 5
elements: Second largest element =
4

ANJALI OVIYA U (192521452)

Occurrence of an element

1 mark

Write a C program to Find the First and Last Occurrence of an Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, x;
5     printf("Enter number of elements: ");
6     scanf("%d", &n);
7
8     int a[n];
9     printf("Enter array elements: ");
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &a[i]);
12    }
13
14    printf("Enter element to search: ");
15    scanf("%d", &x);
16
17    int first = -1, last = -1;
18
19    for (int i = 0; i < n; i++) {
20        if (a[i] == x) {
21            if (first == -1) first = i;
22            last = i;
23        }
24    }
25
26    if (first == -1) {
27        printf("Element %d not found\n", x);
28    } else {
29        printf("First occurrence: %d\n", first);
30        printf("Last occurrence: %d\n", last);
31    }
32 }
```

Input

```
6
1 2 3 2 4 2
2
```

Output

```
Enter number of elements: Enter
array elements: Enter element to
search: First occurrence: 1
Last occurrence: 5
```

ANJALI OVIYA U (192521452)

Q10 Missing Element

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     printf("Enter number of elements: ");
6     scanf("%d", &n);
7
8     int arr[n];
9     printf("Enter %d elements: ", n);
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14    // Assuming array contains numbers from 1 to n+1 with exactly one missing
15    int total = (n + 1) * (n + 2) / 2; // sum of 1..(n+1)
16    int sum = 0;
17
18    for (int i = 0; i < n; i++) {
19        sum += arr[i];
20    }
21
22    int missing = total - sum;
23    printf("Missing element is: %d\n", missing);
24
25    return 0;
26 }
```

 Submitted

Input

```
4
1 2 3 5
```

Output

```
Enter number of elements: Enter 4
elements: Missing element is: 4
```